Explainable Multimodal Organ-Level Fusion for Exploring Systemic Diabetic Effects Across Retina, Kidney, and Heart

Aravindh M[1], Yuvan Nagalingam H[1], Kundan Ratakonda[1] and Shiny Angel T S[1]
[1]*Department of Computational Intelligence, SRM Institute of Science and Technology, Chennai, India*
{*am0021, yh1720, kr0233, shinyant*}*@srmist.edu.in*

Keywords: Chronic Kidney Disease, Diabetes Mellitus, Electrocardiogram (ECG), EfficientNet, Multimodal Analysis, Retinal Fundus Imaging.

Abstract: Diabetes mellitus is a chronic metabolic condition that affects more than 500 million people around the globe, causing progressive impairment in various organs such as the retina, kidney, and cardiovascular system. The current methodologies mostly depend on unimodal techniques that prevent the full comprehension of the disease progression within the whole system. In this research, we present a multimodal fusion framework that utilizes retinal fundus images, renal biomarkers, and ECG data as inputs. EfficientNet models are built for image-based modalities, while XGBoost models are created for tabular renal and cardiac data, in addition to a hybrid CNN-Transformer framework for ECG data. The proposed system is based on score-level fusion with a clear explanation for systemic disease progression.

1 INTRODUCTION

Diabetes mellitus is a chronic metabolic disorder that affects multiple organs, including the retina, kidneys, and cardiovascular system. These complications often develop silently and are detected only at advanced stages, making early systemic assessment essential.

Retinal fundus imaging has emerged as a powerful non-invasive tool for systemic disease analysis, with studies demonstrating its ability to predict cardiovascular risk factors and metabolic biomarkers (Poplin et al., 2017; Rim et al., 2020). Retinal features have also been linked to chronic kidney disease, highlighting cross-organ relationships (Shi et al., 2023; Zhang et al., 2021; Hsu et al., 2025).

Electrocardiogram (ECG) signals have been shown to encode metabolic and autonomic stress patterns associated with diabetes and cardiovascular risk (Sau et al., 2024). Similarly, renal imaging and biomarker-based studies provide early indicators of kidney dysfunction (Betzler et al., 2023).

Despite these advances, most existing approaches remain modality-specific and fail to capture interactions across multiple organs (Tan et al., 2024; Grzybowski et al., 2024; Hamzah et al., 2024).

To address this limitation, this work proposes an explainable multimodal organ-level fusion framework that integrates retinal, renal, and cardiac signals into a unified and interpretable system.

2 RELATED WORK

Retinal fundus imaging has been widely used for systemic disease prediction. Deep learning models have demonstrated strong performance in estimating cardiovascular risk factors and metabolic biomarkers from retinal images (Poplin et al., 2017; Rim et al., 2020). Extensions of this work have explored retina–kidney relationships, showing associations with chronic kidney disease (Shi et al., 2023; Zhang et al., 2021).

Further studies have proposed deep learning frameworks for detecting diabetic kidney disease from retinal images, highlighting the potential of cross-organ prediction (Betzler et al., 2023; Hsu et al., 2025). However, these approaches often suffer from limited interpretability and generalization (Tan et al., 2024).

ECG-based approaches have shown that cardiac electrical activity contains signatures of metabolic dysfunction, enabling detection of diabetes and cardiovascular risk (Sau et al., 2024). However, these models are typically used independently and lack multimodal integration (Grzybowski et al., 2024).

Multimodal approaches combining retinal images with clinical or laboratory data have demonstrated improved performance in disease detection (Bhak et al., 2025). Nevertheless, these methods often rely on tightly coupled datasets and lack reproducibility across different populations (Hamzah et al., 2024).

Overall, existing research lacks a unified, explainable framework that integrates heterogeneous organ-level signals across the retina, kidney, and heart.

3 METHODOLOGY

3.1 Overall System Architecture and Fusion Strategy

The proposed system is designed to explore systemic diabetic effects by integrating organ-level signals from the retina, kidney, and heart through an explainable multimodal fusion framework. The architecture follows a modular, organ-centric design to ensure interpretability, scalability, and reproducibility, as shown in Figure 1.

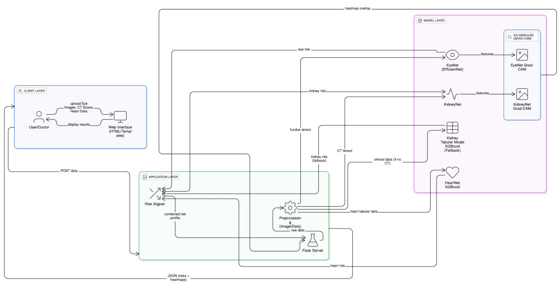


Figure 1: Overview of the multimodal pipeline.

The framework has three main components which operate through three distinct layers. The first layer provides a user interface which enables data entry and visual display through HTML and JavaScript technologies. The second layer uses Flask to manage data preprocessing, inference execution, and result combination.

The third layer contains separate learning modules which study different body systems. Diabetic stress evaluation begins with dedicated pipelines which extract vital information from each organ modality. The extracted data is later combined for complete body system evaluation.

Organ-Level Modeling: Retinal fundus images are processed by an image-based encoder (EyeNet), based on EfficientNet-B3, trained to capture microvascular patterns related to diabetic effects. The model outputs an eye-specific risk score along with spatial explanation maps generated using Grad-CAM.

Kidney-related inputs are handled through two complementary pathways depending on data availability: a kidney imaging encoder (KidneyNet), also based on EfficientNet-B0, for CT-based inputs, and a tabular renal model for clinical biomarker data.

The system uses HeartNet which combines CNN and Transformer models to process multi-lead ECG signals. The convolutional layers extract morphological waveform features, while stacked transformer encoder blocks capture long-range temporal dependencies and electrical stress patterns, enabling robust modeling of cardiac abnormalities associated with systemic metabolic dysfunction.

Multimodal Fusion Modeling: The proposed framework uses a risk alignment method which assigns different weights to various risks. The organ-specific models generate risk scores which correctly measure diabetic stress according to each medical imaging modality.

The application layer first aligns the scores before they undergo fusion processing. The fusion module enables the system to capture both concordant and discordant organ involvement, allowing it to handle cases where diabetes affects different organs in different ways.

3.2 Datasets and Data Sources

The study utilizes publicly available datasets across three modalities. Retinal fundus images are obtained from the EyePACS dataset. Cardiac analysis uses the PTB-XL ECG dataset along with structured clinical data including age, BMI, and HbA1c. Renal analysis uses both kidney CT scan datasets and tabular biomarker datasets including creatinine and eGFR.

3.3 Organ-Level Model Design

Retinal Model (EyeNet): The retinal encoder is based on EfficientNet-B3 and is trained to capture microvascular patterns associated with diabetic stress, as illustrated in Figure 2. It produces a risk score along with Grad-CAM visual explanations.

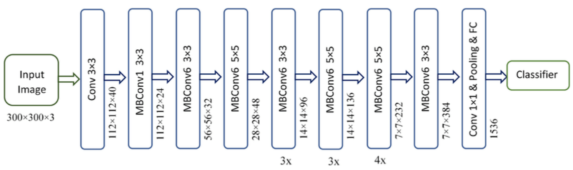


Figure 2: EyeNet model with EfficientNet-B3 backbone.

Renal Model (KidneyNet): The renal module consists of an imaging pathway using EfficientNet-B0 for CT scans and a fallback tabular pathway using XGBoost for biomarker analysis, as shown in Figure 3.

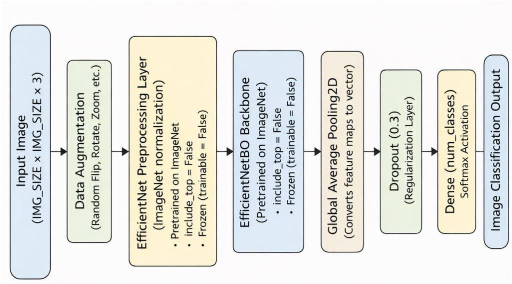


Figure 3: KidneyNet model for CT scan slice analysis.

Cardiac Model (HeartNet): The cardiac module combines CNN and Transformer architectures to process ECG signals, capturing both morphological and temporal features, as depicted in Figure 4.

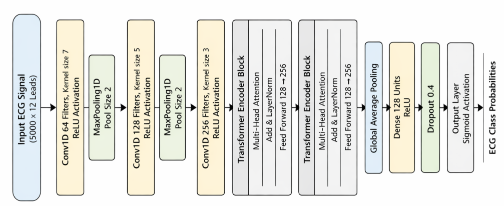


Figure 4: HeartNet model for ECG-based cardiac feature extraction.

3.4 Multimodal Fusion Mechanism

Organ-level models produce normalized risk scores, which are combined using a weighted aggregation:

$$SRS = w_{eye}P_{eye} + w_{kidney}P_{kidney} + w_{heart}P_{heart} \quad (1)$$

where $w_{eye} = 0.4$, $w_{kidney} = 0.3$, and $w_{heart} = 0.3$.

The fusion mechanism captures both concordant and discordant organ involvement, enabling a comprehensive systemic assessment.

3.5 System Output

The final output is a structured systemic risk profile that includes organ-specific scores and visual explanations. Results are delivered through a web-based interface using standardized JSON responses.

4 EXPERIMENTAL SETUP

4.1 Data Preprocessing

Each dataset undergoes modality-specific preprocessing to ensure consistency and improve model performance.

For retinal images, preprocessing includes resizing, normalization, and augmentation techniques such as rotation and horizontal flipping. These steps improve generalization and reduce overfitting.

ECG signals are filtered to remove noise and baseline drift, followed by segmentation into fixed-length sequences. Signal normalization is applied to ensure consistency across samples.

Renal CT images are resized and normalized to maintain uniformity. The tabular data is cleaned, normalized, and encoded where necessary. Missing values are handled using appropriate imputation techniques.

4.2 Training Configuration

All models are trained using the Adam optimizer. The initial learning rate is selected from $\{10^{-3}, 5\times10^{-4}\}$. Early stopping is employed to prevent overfitting, and dropout layers are incorporated within the model architectures.

The dataset is split into training (70%), validation (15%), and testing (15%) subsets. Batch sizes are selected based on modality-specific requirements, with smaller batches used for high-resolution imaging data.

Training is performed in a GPU-enabled environment to accelerate convergence and improve computational efficiency.

4.3 Performance Metrics

Model performance is evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. In addition, the Area Under the Receiver Operating Characteristic Curve (AUC–ROC) is used to assess the model's discriminative ability and separability across classes.

For multimodal analysis, evaluation is performed at both the individual modality level and the combined system level. The Systemic Risk Score (SRS) is used to measure how effectively the model captures interactions across different organs and integrates heterogeneous data sources.

4.4 Technical Specifications

The framework is implemented using Python. Deep learning models are developed using TensorFlow and Keras, while the application layer is built using the Flask web framework to enable real-time inference and integration of multimodal inputs.

All experiments are conducted under consistent settings to ensure reproducibility. Hyperparameters are tuned systematically to optimize model performance across different data modalities.

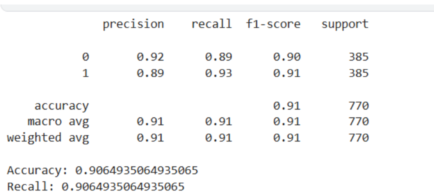
5 RESULTS

5.1 Organ-Level Model Performance

Every organ-specific model was individually evaluated on the basis of common classification metrics, in order to test its capabilities to recognize stress signals related to diabetes in a particular organ.

5.1.1 EyeNet Retinal Model (EfficientNet-B3)

The retinal model achieved an overall accuracy of 90.6%. Figure 5 shows the classification report with balanced performance across classes, with F1-scores of 0.90 and 0.91. Precision (0.92, 0.89) and recall (0.89, 0.93) values indicate stable discrimination without significant class imbalance.



	precision	recall	f1-score	support
0	0.92	0.89	0.90	385
1	0.89	0.93	0.91	385
accuracy			0.91	770
macro avg	0.91	0.91	0.91	770
weighted avg	0.91	0.91	0.91	770

Accuracy: 0.9064935064935065
Recall: 0.9064935064935065

Figure 5: Classification report of EyeNet model.

5.1.2 KidneyNet Renal Model (XGBoost Tabular)

The tabular renal model achieved a test accuracy of 97.5%, with strong predictive reliability on structured biomarkers. Figure 6 shows the ROC curve with near-perfect separability and an AUC of 0.9994.

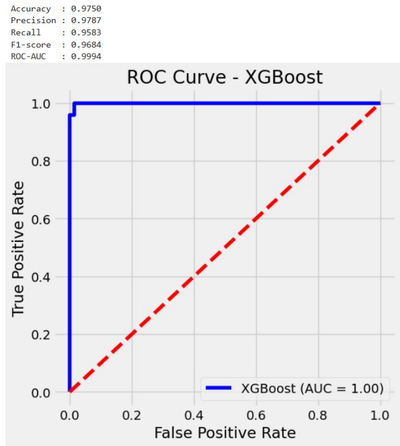
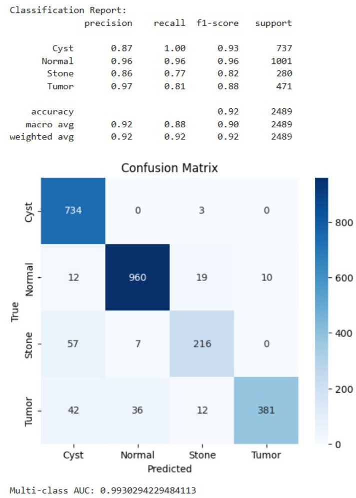


Figure 6: ROC curve of the XGBoost-based renal model.

5.1.3 KidneyNet CT Scan Classification

The CT-based model achieved 92% accuracy across multiple classes. Figure 7 shows the confusion matrix, highlighting strong classification performance, particularly for Cyst and Tumor classes.



Classification Report:

	precision	recall	f1-score	support
Cyst	0.87	1.00	0.93	737
Normal	0.96	0.96	0.96	1001
Stone	0.86	0.77	0.82	280
Tumor	0.97	0.81	0.88	471
accuracy			0.92	2489
macro avg	0.92	0.88	0.90	2489
weighted avg	0.92	0.92	0.92	2489

Figure 7: Confusion matrix for kidney CT classification.

5.1.4 HeartNet ECG-Based Classification

The ECG model demonstrated stable training behavior. Figure 8 shows the training loss and AUC curves. The model achieved an AUC of 0.9186 and a weighted F1-score of 0.76.

Table 1 summarizes the performance across all modalities.

Table 1: Organ-Level Model Performance.

Modality	Acc. (%)	Prec.	Recall	F1	AUC
Retina	90.6	0.91	0.906	0.91	0.96
Kidney	97.5	0.98	0.96	0.97	0.99
Heart	91.86	–	–	0.76	0.9186

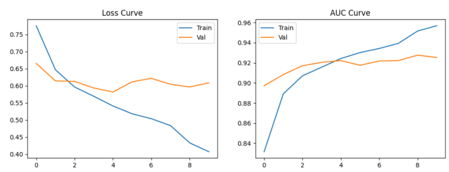


Figure 8: Training loss and AUC curves of ECG model.

5.2 Comparative Analysis

The proposed multimodal framework demonstrates complementary strengths across the three diagnostic pathways. The tabular renal model achieved the highest performance, with near-perfect discrimination (AUC = 0.9994), indicating that structured biomarkers provide a highly separable feature space.

The kidney CT model also showed strong performance (accuracy = 92%, AUC = 0.9930), effectively capturing spatial anatomical features. In contrast, the ECG-based model achieved an AUC of 0.9186, reflecting robust performance despite the inherent complexity of temporal cardiac signals.

Grad-CAM visualizations (Fig. 9) confirm that all models focus on clinically relevant regions, supporting interpretability. Overall, these results highlight the complementary nature of the modalities, with tabular data offering high separability and imaging/ECG enabling scalable and automated screening.

Figure 9: Grad-CAM comparison across retinal, kidney, and ECG models.

5.3 Fusion-Level Systemic Risk Analysis

The fusion module combines organ-level risk scores into a unified systemic representation:

$$SRS = w_{eye}P_{eye} + w_{kidney}P_{kidney} + w_{heart}P_{heart} \quad (2)$$

where $w_{eye} = 0.4$, $w_{kidney} = 0.3$, and $w_{heart} = 0.3$.

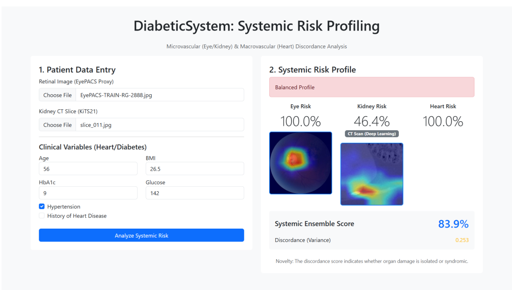


Figure 10: Fusion-level systemic risk visualization.

Figure 10 shows the fusion-level systemic risk visualization, illustrating how organ-level predictions are integrated. Table 2 presents representative patient profiles derived from the fusion mechanism.

Table 2: Representative Patient Profiles Using Fusion Metrics.

Profile	Eye	Kidney	Heart	SRS	Interpretation
Balanced	0.85	0.82	0.80	0.83	Uniform multi-organ stress
Isolated Ocular	0.92	0.15	0.10	0.44	Advanced retinopathy only
Preserved Renal	0.78	0.12	0.75	0.57	Renal resilience present
Healthy	0.10	0.08	0.12	0.10	Low overall risk

The fusion results demonstrate that the model captures both concordant and discordant organ involvement, enabling classification of diverse systemic profiles.

6 DISCUSSION

This study explored systemic diabetic effects through an explainable multimodal organ-level fusion framework integrating retinal, renal, and cardiac signals. The results demonstrate that meaningful diabetes-related patterns can be independently extracted from each organ modality and coherently combined to provide a holistic systemic representation. Importantly, the framework does not rely on patient-matched data or disease-specific diagnosis, but instead emphasizes interpretability and cross-organ understanding over direct clinical prediction.

The organ-level findings highlight the complementary nature of the selected modalities. Retinal fundus images showed strong discriminative capability, consistent with prior evidence that retinal microvascular alterations reflect cumulative metabolic and vascular stress in diabetes (Grzybowski et al., 2024; Tan et al., 2024). Renal models captured structural and functional indicators of kidney involvement, aligning with established links between diabetic kidney disease and imaging or biochemical markers (Betzler et al., 2023; Dai et al., 2021). ECG-based models further demonstrated that cardiac electrical signals encode metabolic and autonomic alterations associated with diabetes, supporting emerging evidence of ECG-derived metabolic signatures (Poplin et al., 2017; Dai et al., 2024).

Multimodal fusion improved system-level performance compared to any single modality, reinforcing the hypothesis that diabetic effects manifest heterogeneously across organs. The fusion strategy moderated discordant organ signals, producing balanced systemic outputs rather than over-reliance on a dominant modality. By preserving organ-wise outputs and providing modality-specific explanations, the framework ensures transparency and reproducibility, consistent with recent recommendations for interpretable multimodal research in chronic disease analysis (Hamzah et al., 2024; Rim et al., 2020).

7 LIMITATIONS AND FUTURE WORK

Despite the promising results, this study has several limitations. First, the datasets used for retinal, renal, and cardiac analysis are not patient-matched, which limits the ability to draw patient-specific conclusions or predict long-term clinical outcomes. Second, the current evaluation is based on a cross-sectional design, restricting the understanding of temporal disease progression. The current score-level fusion approach provides a simplified representation of systemic interactions. While it offers interpretability, more advanced fusion strategies are required to better capture complex cross-modal relationships.

Future work will focus on incorporating patient-matched multimodal datasets to enable longitudinal analysis and improve clinical relevance. The integration of additional data sources, such as laboratory biomarkers and lifestyle factors, is expected to enhance the robustness of systemic representations. Furthermore, advanced fusion techniques will be explored to better model inter-organ dependencies and to generate richer and more informative multimodal representations.

8 CONCLUSION

Diabetes mellitus is a complex metabolic disorder that affects multiple organs, including the retina, kidney, and heart. Conventional diagnostic approaches rely on independent, organ-specific assessments, which limits the understanding of systemic interactions and poses challenges in resource-constrained clinical environments.This work presents an explainable multimodal organ-level fusion framework that integrates retinal, renal, and cardiac data to study systemic diabetic effects. Organ-specific models are developed using EfficientNet-B3 for retinal imaging, EfficientNet-B0 for renal imaging, and a hybrid CNN–Transformer architecture for cardiac analysis. A Flask-based platform is used to combine predictions through a transparent score-level fusion mechanism, supported by visual explanations using Grad-CAM.

Experimental results demonstrate that each modality captures distinct yet complementary information related to diabetes, while their fusion provides a more comprehensive representation of systemic impact. The proposed approach highlights the feasibility of integrating multimodal data for interpretable and scalable diabetic assessment. Overall, this study establishes a foundation for explainable multimodal analysis of systemic diseases and provides a reproducible framework for future research in cross-organ diabetic modeling.

REFERENCES

Betzler, B. K., Chee, E. Y. L., He, F., Lim, C. C., Ho, J., Hamzah, H., Tan, N. C., Liew, G., McKay, G. J., Hogg, R. E., et al. (2023). Deep learning algorithms to detect diabetic kidney disease from retinal photographs in multiethnic populations with diabetes. *Journal of the American Medical Informatics Association*, 30(12):1904–1914.

Bhak, Y., Lee, Y. H., Kim, J., Lee, K., Lee, D., Jang, E. C., Jang, E., Lee, C. S., Kang, E. S., Park, S., et al. (2025). Diagnosis of chronic kidney disease using retinal imaging and urine dipstick data: multimodal deep learning approach. *JMIR Medical Informatics*, 13(1):e55825.

Dai, L., Sheng, B., Chen, T., Wu, Q., Liu, R., Cai, C., Wu, L., Yang, D., Hamzah, H., Liu, Y., et al. (2024). A deep learning system for predicting time to progression of diabetic retinopathy. *Nature medicine*, 30(2):584–594.

Dai, L., Wu, L., Li, H., Cai, C., Wu, Q., Kong, H., Liu, R., Wang, X., Hou, X., Liu, Y., et al. (2021). A deep learning system for detecting diabetic retinopathy across the disease spectrum. *Nature communications*, 12(1):3242.

Grzybowski, A., Jin, K., Zhou, J., Pan, X., Wang, M., Ye, J., and Wong, T. Y. (2024). Retina fundus photograph-based artificial intelligence algorithms in medicine: a systematic review. *Ophthalmology and therapy*, 13(8):2125–2149.

Hamzah, N. A. A., Zaki, W. M. D. W., Halim, W. H. W. A., Mustafar, R., and Saad, A. H. (2024). Evaluating the potential of retinal photography in chronic kidney disease detection: a review. *PeerJ*, 12:e17786.

Hsu, C.-F., Yu, T.-M., Wu, Y.-L., Wang, W.-C., Wang, J.-S., and Chang, S.-S. (2025). Prediction of advanced chronic kidney disease through retinal fundus images by deep learning. *Scientific Reports*, 15(1):37318.

Poplin, R., Varadarajan, A., Blumer, K., Liu, Y., Mcconnell, M., Corrado, G., Peng, L., and Webster, D. (2017). Predicting cardiovascular risk factors from retinal fundus photographs using deep learning. arxiv 2017. *arXiv preprint arXiv:1708.09843*, 10.

Rim, T. H., Lee, G., Kim, Y., Tham, Y.-C., Lee, C. J., Baik, S. J., Kim, Y. A., Yu, M., Deshmukh, M., Lee, B. K., et al. (2020). Prediction of systemic biomarkers from retinal photographs: development and validation of deep-learning algorithms. *The Lancet Digital Health*, 2(10):e526–e536.

Sau, A., Pastika, L., Sieliwonczyk, E., Patlatzoglou, K., Ribeiro, A. H., McGurk, K. A., Zeidaabadi, B., Zhang, H., Macierzanka, K., Mandic, D., et al. (2024). Artificial intelligence-enabled electrocardiogram for mortality and cardiovascular risk estimation: a model development and validation study. *The Lancet Digital Health*, 6(11):e791–e802.

Shi, S., Gao, L., Zhang, J., Zhang, B., Xiao, J., Xu, W., Tian, Y., Ni, L., and Wu, X. (2023). The automatic detection of diabetic kidney disease from retinal vascular parameters combined with clinical variables using artificial intelligence in type-2 diabetes patients. *BMC Medical Informatics and Decision Making*, 23(1):241.

Tan, Y., Ma, Y., Rao, S., and Sun, X. (2024). Performance of deep learning for detection of chronic kidney disease from retinal fundus photographs: A systematic review and meta-analysis. *European Journal of Ophthalmology*, 34(2):502–509.

Zhang, K., Liu, X., Xu, J., Yuan, J., Cai, W., Chen, T., Wang, K., Gao, Y., Nie, S., Xu, X., et al. (2021). Deep-learning models for the detection and incidence prediction of chronic kidney disease and type 2 diabetes from retinal fundus images. *Nature biomedical engineering*, 5(6):533–545.